

Auditing Racial Bias in Healthcare Utilization Prediction

Project Proposal — March 2026

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1. Project Title & Domain

Title: Auditing Racial Bias in Healthcare Utilization Prediction

Domain: Healthcare. The idea is to look at how machine learning models predict which patients will need a lot of healthcare visits in a given year. This matters because if a model is biased, it could lead to some groups getting fewer resources than they actually need. The project is partly inspired by a 2019 study by Obermeyer et al. in *Science* that found racial bias in a major healthcare algorithm used across the US.

2. Dataset Selection

I plan to use the **Medical Expenditure Panel Survey (MEPS)**, specifically Panel 19 from 2015. It is a publicly available dataset from the Agency for Healthcare Research and Quality (AHRQ). It contains patient-level data on healthcare visits, costs, demographics, and health conditions.

Property	Details
Source	AHRQ (meps.ahrq.gov), Panel 19, 2015
Estimated Size	Around 15,000 records after cleaning
Target	High utilization (≥ 10 visits/year), binary
Protected Attribute	RACE (White vs. Non-White)
Other Relevant Features	Age, sex, poverty category, insurance coverage, health conditions

This dataset is commonly used in fairness research and is one of the benchmarks included in IBM's AIF360 toolkit, so there is good documentation available for it.

3. The Ethical Problem

Hypothesis: *A model trained to predict healthcare utilization will likely show racial bias, predicting lower utilization for Non-White patients, not because they are healthier but because they historically had less access to healthcare.*

The core issue is that utilization data is shaped by access barriers. Patients who face financial or systemic barriers simply have fewer recorded visits. A model trained on this data could end up reinforcing the problem by allocating fewer resources to groups that were already underserved. I also want to look at whether features like poverty level or insurance type act as proxies for race, meaning the model could still be biased even if we remove the race feature

entirely.

This connects to several SDAIA AI Ethics Principles, mainly **Fairness & Non-Discrimination** and **Transparency & Explainability**.

4. Technical Stack

The following are the main libraries I intend to use:

Library	Purpose
scikit-learn	Training the ML models (Logistic Regression, Gradient Boosting)
Microsoft Fairlearn	Measuring fairness metrics and applying bias mitigation
SHAP	Explaining model predictions (feature importance, individual explanations)
pandas, matplotlib	Data processing and visualization

5. Preliminary Methodology

The general plan is to go through the following steps:

Data Preparation:

Load and preprocess the MEPS dataset. Create the binary race and utilization variables.
Select relevant features and split into training and test sets.

Model Training:

Train at least two models: a Logistic Regression as a simple baseline and a Gradient Boosting Classifier as a stronger model. Compare their overall performance.

Fairness Audit:

Use Fairlearn to measure how the models perform across racial groups. The main metrics I plan to look at are Demographic Parity Difference, Equalized Odds Difference, and the per-group false negative rate.

Explainability:

Use SHAP to understand which features are driving the model's decisions, and whether race or race-correlated features play a significant role.

Mitigation:

Try to reduce the bias using Fairlearn's mitigation algorithms, such as ThresholdOptimizer or ExponentiatedGradient. Then compare the mitigated model against the original to see how much fairness improves and what accuracy cost there is.